FinCaKG: A Framework to Construct Financial Causality Knowledge Graph from Text*

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Causality analysis holds a prominent role in finance, and the presentation of causality could offer valuable insights for risk mitigation, investment decisions, and portfolio optimization. Recent research has extensively investigated the identification of causality from text, yet there is still a significant deficiency in providing a comprehensive causality presentation from those textual discoveries. In this paper, we present an end-to-end framework to automatically construct Financial Causality Knowledge Graph (FinCaKG) from text, which allows us to visualize the captured causality from a holistic perspective. This framework involves three distinct tasks, including causality sentence detection, cause-effect span identification, and causal dependency representation. To examine the adaptability of FinCaKG framework, we generate, compare, and analyze the distinct FinCaKGs created from different corpus. The results show that this framework has the capability to not only capture the confidential causality but also represent them in a highly detailed manner in the resulting knowledge graph - FinCaKG. We perform a comparative study with ConceptNet, revealing the notable contributions of FinCaKGs in terms of domain coverage and the density of its causal knowledge. Furthermore, we implement two case study to examine the capability of financial role declaration and distinction of ChatGPT and FinCaKGs applications. We found that, compared to ChatGPT, FinCaKGs are able to represent the much clearer and more distinctive logic opinions among diverse financial roles.

Keywords: Knowledge representation; Causality; Finance

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1. Introduction

Current knowledge graphs are prone to incorporate plenty of triples to grow larger. However, the volume of knowledge graph does not mean the volume of knowledge, especially expert knowledge (aka. expertise). Many domain-oriented knowledge graphs have been created to present the concepts of expertise. But in general, concepts are linked with too many types of relations, which introduces challenges in establishing meaningful connections between them. To bridge this deficiency, we resort to applying solely causal relation as the linkage and connect the domain keywords if the causality exists between them. We expect that the connections with causality qualify to present the inner logic for a specific domain.

Causal relations have been studied and stored in many open-source knowledge bases, e.g. WikiData [2], ConceptNet [3]. Even the causality-dedicated knowledge graphs have been generated by many works recently, e.g. CausalNet [4], Cause Effect Graph [5], CauseNet [6] and ATOMIC [7]. However, the causal relationships within these graphs are depicted in a fragmented manner, where interconnections between tuples are missing. Due to this chaos, they are prone to ignore the specialty of causality — structuring knowledge into a much more logical and understandable presentation.

In finance, the presentation of causality could offer valuable insights for risk mitigation, investment decisions, and portfolio optimization. Therefore, we narrowed down our investigation to the finance domain for our initial trial, with the prospect of conducting more in-depth research in the future.

In this paper, we present a framework to automatically construct the **F**inancial **C**ausality **K**nowledge **G**raph (**FinCaKG**) from financial reports. We collect the causal and effect span pairs from sentences by implementing a KG-embedded BERT model. To strengthen causality linkage in a graph structure, we propose to transform the pairs of causal spans into structural relations. To facilitate the straightforward visualization of FinCaKG, we provide screenshots to show the properties of nodes and relations and present the cause-effect paths of keywords for expertise discovery. To study the adaptability of FinCaKG framework, in addition to the original FinCaKG version, we even implemented this framework on other corpora and generated two other versions of FinCaKG. The comparison between distinct versions of FinCaKG demonstrates that this framework is capable of uncovering different expertise perspectives on the same concept. Furthermore, we implement a case study to examine the capability of financial role declaration of ChatGPT and FinCaKGs applications. We found that, compared to ChatGPT, FinCaKGs are able to represent the much clearer and more distinctive logic opinions among diverse financial roles. To evaluate the assistance of FinCaKGs in the downstream text generation task, we assess the readability and comprehensibility of the generated text with and without FinCaKGs. The inclusion of FinCaKGs highlights the requirement for readers to possess a greater depth of knowledge in relevant domains, suggesting the need for more extensive educational background to facilitate understanding.

2. Related Work

Pattern-based approach: The earliest approaches of causality detection relied on finding the explicit causality in sentences. From summarizing the idiomatic causality, experts could generate various handcrafted rules with syntactic, lexical, and grammatical patterns [4, 8–10].

Shallow machine learning approach: Machine learning-based approaches incorporate the causality attributes into feature vectors using both semantic and syntactic characteristics and then apply algorithms to interpret features in different manners, e.g. regression, classification, and clustering [11]. In general, we select the causality features by pattern-based approaches and use machine learning approaches to capture the implicit causality [12, 13].

Deep learning approach: Getting rid of the crafty procedures at feature engineering, deep learning approaches can learn features in training automatically. However, for causality relation, the performance of popular RNN [14] or CNN [15] is limited. Therefore, more specialized models were examined and explored [16]. The typical LSTM [17] allows the past elements to be considered as context for an element under scrutiny. Additionally, considering the future sequence context, bidirectional LSTMs (Bi-LSTM) [18] were demonstrated to outperform in identifying cause-effect from text [19].

Causality knowledge graph construction: In the text, the straightforward causality is the pairwise cause-effect elements. It can be word pairs, i.e. *bacteria—disease*, or it could be textual span pairs. i.e. *Because of their wrong investment—they went bankrupt.* From scratch, some work collected abundant causal word pairs and assigned causal relations with weighted occurrences, e.g. Cause Effect Graph [5] and CauseNet [6]. Frattini et al. investigated the statistics of causality in text and they showed that textual span pairs have greater quantity than word pairs [20]. Additionally, textual span pairs are capable of expressing more concrete causality. However, taking advantage of causal span pairs to construct a knowledge graph remains blank. Therefore, we identify causal span pairs from the text and propose a methodology to exploit span pair connections to generate a causal knowledge graph.

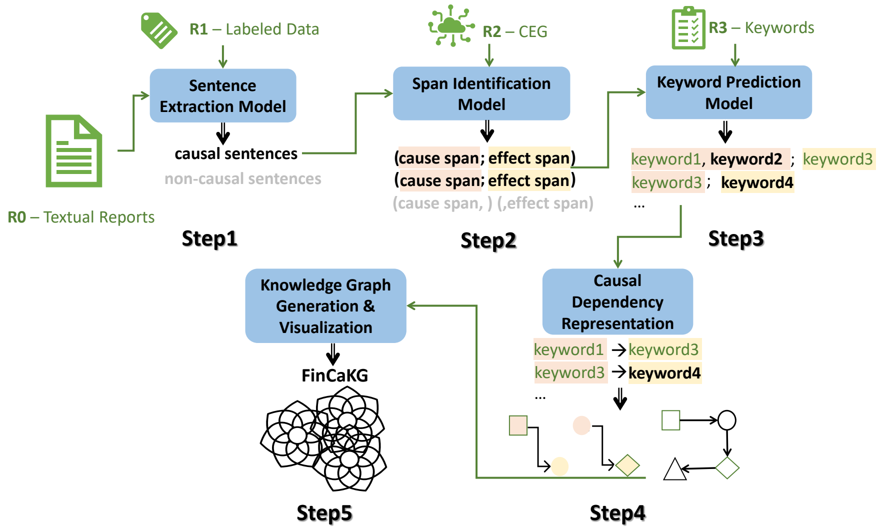


Fig. 1: The framework of FinCaKG construction.

3. FinCaKG Framework

We strive to automatically construct FinCaKG from raw text while maintaining the expertise and trustworthiness of FincaKG. In pursuit of this objective, we segment the entire workflow into a series of consecutive steps. Each step is dedicated to a specific task and utilizes its own set of supporting resources. This approach allows us to ensure that the performance of each step is both quantifiable and traceable. Correspondingly, we draw the entire framework of constructing FinCaKG step by step as shown in Figure 1:

- Step1: extract causal sentences
- Step2: identify cause-effect textual spans from causal sentences
- Step3: predict financial keywords of cause-effect spans
- Step4: formulate the keyword causality dependency from cause-effect spans
- Step5: generate and visualize FinCaKG

In the procedures of dealing with text (Step1-3), we mainly exploit the diverse capabilities of language models in the tasks of sentence classification (Step1) and token classification (Step2-3). To assist with training on those models, we also list the resources of labeled data and external knowledge at the top of the corresponding workflow.

Particularly, because the identification of textual spans (Step3) plays a pivotal role in uncovering causality within this framework, we will delve into a detailed discussion of the models we employ for this specific task. During FinCaKG construction (Step4), we propose to connect the keywords pairs with causality dependency from tail to head, to establish interconnected relationships in the knowledge graph representation.

3.1. *Components of Models*

In the current research, many tasks have been extensively explored, e.g. *huggingface*, each model has implemented predefined architectures for different tasks, including sequence classification and token classification tasks. The sentence extraction model (Step1), suits well the sequence classification task. Those baseline models are quite competent to classify all the text into causal sentences or non-causal sentences. It is the same situation as the keywords prediction model (Step2), which is considered a token classification task with the BIO scheme. The results of our experimental assessment in Section 4.2 substantiate the effectiveness of baseline models.

In contrast, the cause-effect span identification model (Step2) remains a critical and complex issue, primarily due to its subpar performance from its relative baseline models. Moreover, in the causal dependency representation model (Step4), we put forward a fresh perspective of converting textual associations into structural causality within a graph representation. This particular step becomes significant as it involves consolidating the findings from earlier models and laying the groundwork for establishing causal dependencies during the construction of FinCaKG.

3.2. *Span Identification Model*

Given the sentence *"Zhao found himself 60 million yuan indebted after losing 9,000 BTC in a single day (February 10, 2014)"*, we can identify *"losing 9,000 BTC in a single day (February 10, 2014)"* as the cause while we annotate *"Zhao found himself 60 million yuan indebted)"* as the effect.

Recently, many methods have been proposed in the FinCausal workshop [21, 22]. We noticed a highlighted model, DTGNN [23], the winner of FinCausal 2021, who incorporated dependency relation features from a sentence through a graph neural network into a token classifier with Viterbi decoding [24]. This system mainly focuses on adding syntactic features by the dependency features. However, once the syntactic clues are absent in the text, this model sometimes reverses the cause and effect spans. To solve this problem, our previous model, namely **ilab-FinCau** [25], introduced a graph construction technique to inject cause-effect graph knowledge into graph embedding. The graph features combined with BERT embedding are used to predict the cause and effect spans. We resort to utilizing this proposal as our cause-effect span identification model and any explanations of the forthcoming model refer to this algorithm.

The cause-effect span identification model is composed of different functional modules: BERT encoder, graph builder, GNN+BiLSTM, and Viterbi Decoder. During the graph builder module, this model inserts the causality knowledge from Cause-Effect Graph(CEG) [5] into embeddings using graph neural networks. The results show the proposed graph builder method outperforms other competitive

Table 1: The statistics of resources applied in relative steps.

Steps	**Units**	**Train**	**Prediction**
Step1	#sents (#causal sents)	$14k(1k)$	$10m(525k)$
Step2	#sents	$2.7k$	$525k$
	#sents after postprocessing	-	$497k$
Step3	#sents	$497k \times \frac{9}{10}$	$497k \times \frac{1}{10}$
	#unique investopedia NPs	1317	1317
	#unique new financial NPs	-	1623

models with or without external knowledge. For more technical details of **ilab-FinCau** [25], please refer to the original paper.

3.3. *Causal Dependency Representation*

Causal span pairs are not as easily handy as causal word pairs in the graph structure. In the preparation before construction, we seek solutions to transform span pairs into word pairs according to the following hypotheses in semantics:

- Given a sentence, a domain-specific concept conveys more information than general contextual terms.
- If two sentences have a relation, in general, the core concepts of each sentence possess the same relation in semantics.

Based on the aforementioned hypotheses, the connections of the domain-specific **N**oun **P**hrases (NPs) of cause span to that of effect span can be considered causal relations. With the theoretical support, the gist of this kind of transformation is to retrieve the domain-specific (i.e. finance) NPs of the target spans. Here we simplified the financial NPs as Keywords. As shown in Step3 of Figure 1, the bold keywords are predicted as the new financial terms while the green keywords are from the known lists. Then in Step4, the tagged keywords of each sentence are connected in pairs, from cause spans to effect spans. In the example of Step3 in Figure 1, *keyword3* appears at both cause spans and effect spans in different sentences, and then this keyword is connected as the mediator to link two pairs together. In this manner, causal chains are generated collectively.

To progress through this framework, a complete FinCaKG could be built automatically with enriched connections from the given text.

4. FinCaKG Generation

4.1. *Data Resources*

4.1.1. *Financial 10-k Reports (R0)*

In finance, the financial reports of listed companies are open to the public, where 10-k is one of the formats of the financial report in S.E.C. We focus on the last 5 years' reports of the top 2000 companies of Russell 2000 Index published in 2021 in the last 5 years. In order to directly acquire the machine-readable text, we only deal with the reports in the xbrl extension, for which 5093 reports are left. During preprocessing, we truncate the textual contents into sequential sentences, while eliminating sentences that have string lengths less than 70. Finally, we have around 10 million sentences to be predicted in Step1. Correspondingly, this value is shown in the last column of Step1 in Table 1.

4.1.2. *Labeled Data (R1)*

In the shared tasks of FinCausal[a], they provide the labels for sentences from financial reports. We merge their dataset[b] from different years and deploy them on Step1 and Step2 separately. In Table 1, the column of Train records the statistics of labeled data for Step1 and Step2.

4.1.3. *Causal Word Pairs - CEG (R2)*

Cause **E**ffect **G**raph [5] extracts causal word pairs following the similar strategies as CausalNet [4], but it is developed from a larger and cleaner corpus [26]. It contains 89.1 million causal edges than 13.3 million of CausalNet. We expect to benefit from the numerous causal word pairs in the detection of the causal span pairs. To prepare knowledge embedding in Step2, we identify all the word pairs from CEG that exist in each given sentence. In brief, each sentence matches 3.1 word pairs from CEG on average.

4.1.4. *Financial Keywords (R3)*

Investopdia is famous as a dictionary for financial concepts. Accordingly, we consider its vocabulary[c] as financial keywords, which contains 6259 unique NPs to assist for Step3. However, not all words in the vocabulary are shown in our datasets. As we present in Table 1, only 1317 unique NPs are identified as financial keywords in Step3.

4.2. ***Experiments***

We utilize the aforementioned models of Section 3 that are suitable for steps 1 to 3 based on their performance assessment in comparison to the existing baseline research. We summarize the parameters of models in Table 2. In Step1, we recognize causal sentences as label **1** and other sentences as label **0**. Sequentially, Step2 handles the discovered causal sentences, and it tags each token with one of the five

[a] `http://wp.lancs.ac.uk/cfie/fincausal2022/`, access date: 2022/12/18
[b] `https://github.com/yseop/YseopLab`, access date: 2022/12/18
[c] `https://www.investopedia.com/financial-term-dictionary-4769738`, access date: 2022/12/18

Table 2: The parameters of models in each step.

Steps	**labels**	**models**	**ep.**	**bt.**	**lr.**	**opt.**
Step1	0, 1	BERT	10	8	2e-5	epsilon =1e-8
Step2	B-C, I-C, B-E, I-E, O	ilab-FinCau [25]	5	4	5e-5	
Step3	B-F, I-F, O	XLM-Roberta	5	8	2e-5	

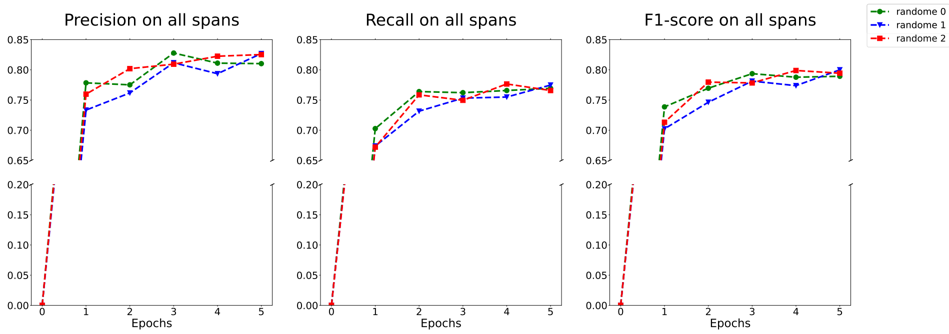
Table 3: The performance of causal sentences extraction model in Step1.

	Dataset	**Precision**	**Recall**	**F1-score**
Validation	Labeld Data (R1)	96.31%	96.07%	96.19%
Prediction	Financial Reports (R0)	95.93%	95.89%	95.91%

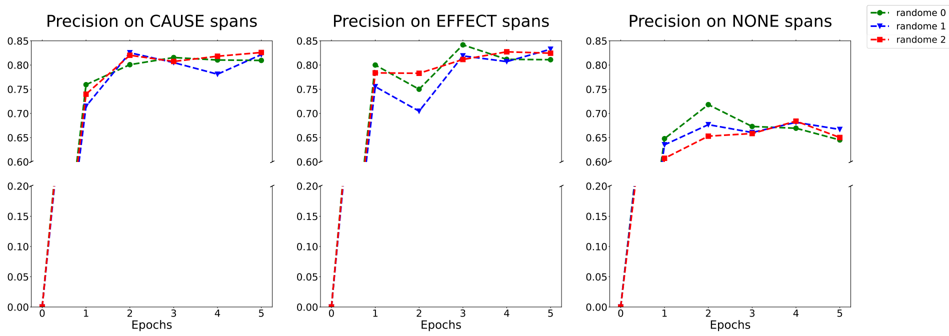
labels (in the label column of Table 2). For instance, a cause span is identified from the sequential labels, i.e. B-**C**, I-**C**, ..., I-**C** for cause part, similar labeling strategy to effect parts with B-**E** and I-**E**. For the rest of the tokens, we tag O denoting none labels. Correspondingly, Step3 follows the same architecture but applies with the other three labels on model XLM-Roberta [27]. B-**F**, I-**F** signify the Beginning and Inner label of financial keywords, and label O is used to tag the rest of the tokens in a sentence. In the rest of the columns, *eps., bt., lr., opt.* depict model parameters with the number of epochs, the number of batches, the learning rate, and the hyperparameter value of the optimizer.

4.2.1. *Performance on causal sentence extraction*

We aim to find causal sentences with high purity from the text. To achieve it, we need to: 1) evaluate the training models on Labeled Data (R1) - shown in *validation* row of Table 3; 2) select the best-trained models to predict financial reports (R0) - shown in *prediction* row of Table 3. We present the precision, recall, and F1-score of binary classification on the causal sentence extraction task in Table 3. In fact, within 10 epochs, we can only observe a slight increase after 1st epoch. Here we show all the results of the validation process in the 10-th epochs. It shows that in *validation* process, this model achieves rather high scores around 0.96. Nevertheless, we select the final trained models and use them for the prediction task. The prediction process yields somewhat lower scores than the validation process, yet they are still deemed



(a) The overall evaluation for all identified textual spans.



(b) The independent evaluation of CAUSE, EFFECT and NONE spans.

Fig. 2: The performance of cause-effect span identification model in Step2.

of sufficient quality for identifying causal sentences.

4.2.2. *Performance on cause-effect span identification*

The target here is to find the cause span and effect span given a causal sentence. To evaluate the prediction of the complete spans in a sentence, including both the boundaries and the label, we apply the sequence labeling metrics - **seqeval**[d]. To mitigate the impact of random variation in model performance, we perform the experiments using the same model with three distinct random states, tagged as *random 0, random 1, random 2* lines in the following figures. The overall evaluation for all identified textual spans is shown in Figure 2a. We observe the increasing scores along with larger epochs, and the general trends of different random states are similar. It suggests that the model keeps learning the features until around 0.8 in the f1-score. Meanwhile, in Figure 2b, we also present the independent evaluation

[d]`https://github.com/chakki-works/seqeval`

Table 4: The performance of financial keywords prediction model in Step3.

10-folds	**Precision**	**Recall**	**F1-score**
avg.	98.72%	98.68%	98.70%
min.	98.10%	98.14%	98.12%
max.	99.07%	99.05%	99.06%

of CAUSE span, EFFECT span and NONE span. The CAUSE span subgraph is calculated collectively with the entire sequential labels for a cause span, and the EFFECT span and None span subgraph are similar correspondingly.

Particularly we notice, for each epoch, if two kinds of spans present increasing trends, the rest span will show a decreasing trend. It is consistent with our observations: one sentence has a fixed number of tokens, and the more tokens are assigned for one span, the fewer tokens for others. In the comparison, we recognize that the cause/effect span supports the entire increasing trend in the first row, while *None* spans present fluctuated curves but end with similar scores as initial ones. It implies that the model is mainly learning to distinguish the cause span from the effect span, which is a desirable finding for this knowledge-embedded model.

4.2.3. *Performance on financial keyword prediction*

We intend to predict the financial keywords among the given text, and those keywords should be as expressive and technical as those of Investopedia NPs. In this case, we plan to consider Investopedia NPs that exist in the corpus (R0) as the ground truth in training text, and accordingly, train the model to predict similar keywords in unseen text. To walk through every section of the textual reports, we deploy the 10-fold cross-validation strategy, where 9/10 reports are used for model training and validation, and the rest 1/10 reports perform as the unseen text for model prediction. To avoid test data leakage, we train models separately and index them by the number of the initial section.

To evaluate the models' performance, we apply the sequence labeling metrics similar to Step2. Collectively, we provide the average, minimum, and maximum values for each model for different runs in 10 folds. As shown in Table 4, we observe the best performances (approximately 0.99) in all metrics. This observation shows that our models remember well which are terms of ground truth and which are not. Besides, we also want to investigate whether those models have sufficient prediction capability on financial keywords. In statistics of Table 1, we notice not all Investopedia NPs (i.e. only 1317) exist in the corpus, while fortunately, the trained models succeed in predicting new 1623 financial keywords.

Table 5: The statistical comparison between distinct versions of FinCaKGs.

	FinCaKG-FR	**FinCaKG-ECT**	**FinCaKG-AR**	**ConceptNet-Cause**[e]
# nodes	1,717	546	199	12,832
# rels	11,633	1,802	283	16,801
# docs	5,093	5,547	10,057	-

5. Evaluation

5.1. *Comparison of distinct FinCaKGs*

We aim to explore the generalization of our suggested framework. Therefore, using the identical FinCaKG methodology, we create diverse FinCaKG versions sourced from various corpora. For instance, in this paper, we introduce financial reports as our main corpus and construct FinCaKG as **FinCaKG-FR**.

Instead, we manage to apply with other two important corpora: Earning Call Transcripts and Analyst Reports. The **Earning Call Transcripts (ECT)**[f] are textual records of conference calls held by publicly traded companies with their investors and analysts. Their contents include questions from analysts and answers from company representatives. The **Analyst Reports (AR)**[g] refer to documents produced by financial analysts who study and analyze publicly traded companies. Their content helps investors make informed decisions about buying, selling, or holding stocks and other financial instruments.

As shown in Table 5, we report the number of documents for each corpus and the number of nodes and relations in the corresponding version of FinCaKGs. For the number of documents, we include 5547 transcripts in 2021 for ECT corpus and, for AR corpus, 10,057 abstracts of analyst reports from 2014 to 2020. Correspondingly, we obtain **FinCaKG-ECT** and **FinCaKG-AR** to compare with our initial **FinCaKG-FR**.

Besides, we also compare the competence of diverse FinCaKGs in terms of expertise discovery. By querying with the same keyword, we obtain many related cause-effects nodes. Due to the different number of nodes in different versions, we cannot guarantee the neighbors of searched keywords are sufficient. Therefore, we show the relative paths of keywords in FinCaKGs, while the keyword is not necessary to be the center of all neighbors.

Please check the contents in Figure 3. On one hand, in subgraph 3a, we show all possible causes and effects after the keyword. Even though only a limited number of nodes are shown up, this subgraph still can demonstrate that *bad debt* mainly influences *operating profit* and *operating expense*. It implies that this opinion or

[e] To preserve space, we also incorporate the Conceptnet-Cause statistics to assist the analysis in Section 5.3.
[f] `https://seekingalpha.com/earnings/earnings-call-transcripts`
[g] `https://www.bloomberg.com/professional/solution/bloomberg-terminal/`

(a) The relative 2-hop neighbors in **FinCaKG-ECT**.

(b) The relative 2-hop neighbors in **FinCaKG-AR**.

Fig. 3: The comparison between diverse FinCaKGs in expertise discovery.

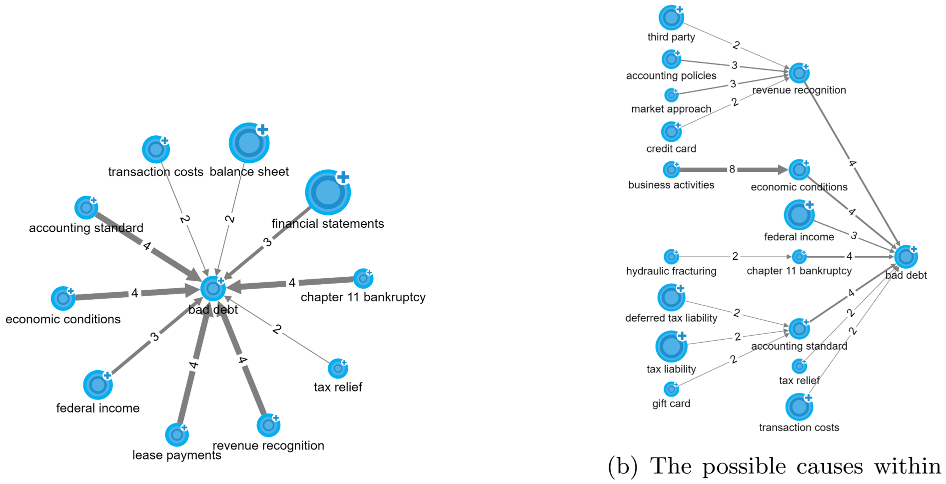
logic path is widely used by company representatives to answer the questions of their investors.

On the other hand, in subgraph 3b, we notice that FinCaKG-AR prone to provide a general and wide horizon to display the effects and causes of *gross margin*, in which *bad debt* is one of the minor reasons. It claims that *supply chain*, *fixed cost* and *market share* might be the major reasons. It explicitly depicts the point of view of analyst reporters during their inferences.

5.2. *Expertise Discovery in FinCaKG*

We have confidence that FinCaKG can unveil expertise by uncovering intricate causal chains. For the sake of a readable view, we start to explore our knowledge graph by querying a financial keyword, i.e. *bad debt*. We expect to seek the possible causes/effects of this keyword. In Figure 4a, we obtain the possible causes from the 1-hop neighbor nodes of this query term. The value of edges is noted by the number of causal pair occurrences in the source financial reports. This subgraph suggests that factors like *accounting standard, economic conditions, lease payments, revenue recognition, bankruptcy,* etc. might be the main reasons to trigger *bad debt*. In the meantime, there are also other minor reasons listed with lower occurrence value, like *transaction costs, balance sheet, financial statements, federal income, and tax relief*.

Moreover, we consider some nodes as intermedia from the 1st-hop nodes and seek the 2nd-hop neighbor nodes if exist. Consequently, in Figure 4b we have many possible causes within 2 hops of *bad debt*. We notice the 2nd-hop nodes following by



(a) The possible causes within 1 hop.

(b) The possible causes within 2 hops.

Notes: A larger node size signifies that this node has connections with a greater number of nodes, that are not displayed in this context.

Fig. 4: The expertise discovery of **FinCaKG-FR** based on financial reports.

revenue recognition, economic conditions, bankruptcy, accounting standard are quite interesting, which mainly talk about the in-depth reasons from different perspectives that cause "bad debt". These paths imply our FinCaKG is capable of digging out fine-grained factors and presenting them in tiers.

5.3. *ConceptNet V.S. FinCaKGs*

To demonstrate the significance and originality of FinCaKGs, we conduct a comparative analysis with ConceptNet [3], a well-established knowledge base that incorporates crowd-sourced knowledge from sources like DBPedia, WordNet, Wiktionary, and more. In order to maintain fairness, we exclusively focus on nodes connected through causality in ConceptNet, and we refer to this subset as **ConceptNet-Cause**.

As shown in Table 5, ConceptNet-Cause boasts roughly six times the number of nodes compared to FinCaKG-FR, but it falls short in terms of the number of relationships. For more insight, we examine the connectivity and domain coverage of ConceptNet-Cause and FinCaKGs in Table 6. Our initial assessment focuses on **density**, which ranges from 0 for a graph with no edges to 1 for a fully connected graph. Then we calculate the average number of direct neighbors for a target node, which we refer to as **node degree**. Of greater significance, we measure the finance **domain coverage** by conducting a fuzzy string matching between nodes and Investopedia vocabulary.

Table 6: The investigation on connectivity and domain coverage of FinCaKGs and ConceptNet-Cause.

Causal KGs	**Density**	**Node Degree**	**Domain Coverage**
ConceptNet-Cause	0.01%	2.62	0.72%
FinCaKG-FR	0.39%	<u>13.55</u>	60.51%
FinCaKG-ECT	0.61%	6.60	<u>83.88%</u>
FinCaKG-AR	<u>0.72%</u>	2.84	83.42%

We conclude that ConceptNet-Cause has a notably low density, reflecting the scarcity of its causal knowledge, and it shows a clear deficiency in finance domain coverage. When it comes to representing financial causal knowledge, existing knowledge bases aren't sufficient. Generating specialized knowledge graphs through the dedicated framework is imperative.

6. Case Study I: Financial Role-Play Showdown

Our financial scopes comprises three specific roles: CEO, accountant, and analyst. The FinCaKGs are intentionally crafted to delineate and individually capture the logical pathways associated with each of these roles. Similarity, our objective is to assess ChatGPT's efficacy in representing different financial roles, emphasizing the clear differences in reasoning among them. Our experiments are designed to prompt ChatGPT to delineate financial roles. Furthermore, beyond assessing its capacity to represent these roles distinctly, we seek to understand ChatGPT's proficiency in distinguishing among the roles from their unique viewpoints. For this purpose, we plan to execute a multi-class classification task with ChatGPT to distinct the financial roles across different viewpoints. The viewpoints could be derived from the generated text or extracted from FinCaKGs. Through the analysis of experiment findings, we endeavor to investigate the current landscape and difficulties encountered by ChatGPT in the domain of financial Role-Play.

6.1. *Financial Roles Declaration*

Assume that each FinCaKG encapsulates the perspectives of the authors, it becomes evident that every FinCaKG embodies a financial role within its corpus. Specifically, this mapping encompasses FinCaKG-ECT for the role of CEO (P1), FinCaKG-FR for the position of accountant (P2), and FinCaKG-AR for the role of analyst (P3). In assessing ChatGPT's capacity to embody financial roles, we focus on the presentation of top-k viewpoints reasoned by different characters for the same question. For the prompt engineering, in the system setting, we describe the target role and the related scenario for question answer task. In the user setting, we formulate inquiries to elicit targeted reasoning, such as exploring 'cause' or 'effect', by presenting prompts like *"How do you explain the possible causes and effects of*

Table 7: The system contents for the prompts of different financial roles.

Financial Roles	System Contents
P1-CEO	*Assume that you are a CEO of a company and you need to reply questions in the Earning Call scenario.* + [response formatting]
P2-Accountant	*Assume that you are a accountant of a company and you need to reply questions based on 10-K report content.* + [response formatting]
P3-Analyst	*Assume that you are an analyst of a third-party financial services company and you need to reply questions to meet the needs of your clients.* + [response formatting]

Prompts for [response formatting]: *Your answer should format as top-k bullet points without explanation and ONLY reply in json format with order as key and opinion points as value.*

Table 8: The averaged similarity score of different roles on the top-k opinions during 'cause' or 'effect' reasoning.

		'cause' reasoning			**'effect' reasoning**		
Models	**Hits@k**	**P1:P2**	**P1:P3**	**P2:P3**	**P1:P2**	**P1:P3**	**P2:P3**
GPT-3.5-turbo	1	0.33	1.00	0.00	0.33	0.00	1.00
	3	0.67	0.89	0.56	0.67	0.78	0.56
	10	0.87	0.67	0.63	0.70	0.83	0.57
GPT-4	1	0.33	1.00	0.00	0.67	1.00	1.00
	3	0.89	1.00	0.78	0.44	0.78	0.67
	10	0.70	0.83	0.57	0.63	0.67	0.37

*Notes: **P1**-CEO; **P2**-Accountant; **P3**-Analyst*

bad debts?". Further details can be found in the prompt descriptions provided in Table 7.

In the experimental setup, we will request the top-k opinions from ChatGPT and evaluate the similarity scores of these opinions across different pairs of roles. The outcomes will be presented in Table 8, with the **Hits@k** column indicating that the comparison is limited to the top-k opinions. Our analysis aims to address two primary questions: 1) Does ChatGPT demonstrate the ability to distinguish notable differences in reasoning among distinct characters? 2) What level of agreement is observed between different versions of ChatGPT?

For the first question, we aim to analyze the similarity levels for the most and least similar pairs of roles. Focusing on the top-1 and top-3 opinions, the **P1:P3** column reveals a high overlap of top opinions, indicating that the most similar pairs are P1-CEO and P3-Analyst. Conversely, for the least similar pair, we focus the lowest values in the range of Top-10, where **P2:P3** column achieve comparatively lowest similarity than other pairs. This suggests that while ChatGPT effectively captures the differences between P2-Accountant and P3-Analyst, it struggles to

Table 9: The performance of financial role distinctions based on GPT-4 and FinCaKGs.

Models	**Role Opinions from:**			
	Generation*	**FinCaKGs**	**Generation***	**FinCaKGs**
	macro-accuracy		**micro-accuracy**	
GPT-3.5-turbo	0.00%	17.00%	19.44%	27.78%
GPT-4	8.33%	17.00%	25.00%	61.11%

Generation: we use previous role declaration of GPT-4.*

differentiate the disparities between P1-CEO and P3-Analyst.

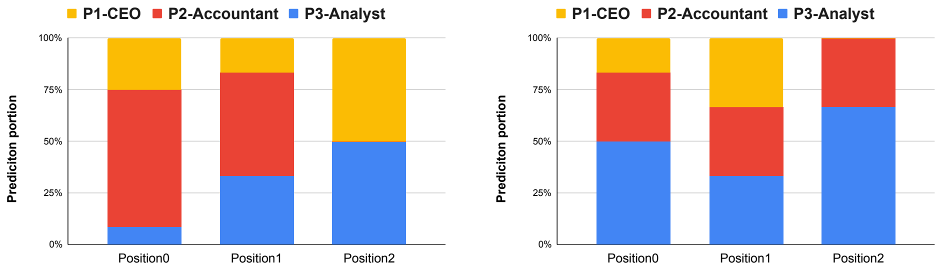
Regarding the second question, our examination of different model showed a roughly consistent similarity between different roles during 'cause' reasoning. However, for 'effect' reasoning, we observed a lower agreement between P1 and P2, particularly noticeable in the top-1 and top-3 opinions.

Concisely, while ChatGPT effectively declare the roles of P2-Accountant and P3-Analyst, it encounters difficulty in distinguishing between P1-CEO and P3-Analyst. For different version of GPTs, they tend to present similar trends in 'cause' reasoning but not 'effect' reasoning.

6.2. *Financial Roles Distinction*

Given the financial role declarations, we proceed to examine whether ChatGPT can accurately classify the self-generated declarations according to their respective role labels. To enhance our evaluation of ChatGPT's discriminative capabilities, we have designed a multi-class classification task for each prompt. Within each prompt, we present a opinion chunk containing the top-k opinions for each role, then we shuffle those opinion chunks into a new sequence. For instance, with three opinion chunks representing three roles, there are six permutations ($P(3, 2) = 6$) for the combination of chunk positions. Each permutation of the three opinion chunks will serve as system content. As for the user content of the prompt, we instruct ChatGPT to assign the labels of the three roles for each position. For evaluation, a response is deemed macro positive only if all labels are correctly assigned; otherwise, it is considered negative. In cases where only partial labels are assigned correctly, the correctly labeled ones are regarded as micro positive, otherwise micro negative.

In our experiments, we consider two opinions resources, i.e. role declaration from GPT-4 and from FinCaKGs. Given the incompleteness of FinCaKGs, we integrate opinions derived from both 'cause' and 'effect' reasoning to match the quantity of opinions from GPT-4. The outcomes, detailed in Table 9, highlight distinct patterns. From both resources of role opinions, the models exhibit notably low macro-accuracy scores, with marginal enhancements in micro-accuracy. This underscores ChatGPT's limitations in role distinction. Notably, GPT-4 demonstrates



(a) Role Opinions from generations

(b) Role Opinions from FinCaKGs

Fig. 5: The position bias in the task of role distinction prediction.

superior performance than GPT-3.5-turbo. It's worth mentioning that, compared to ChatGPT-generated role declarations, FinCaKGs demonstrate higher accuracy values, suggesting greater ease in distinguishing and classifying roles. In other words, ChatGPT displays a significant challenge in role confusion for its self-generated content.

Meanwhile, our investigation aims to determine whether ChatGPT's prediction results exhibit position sensitivity, i.e. consistently assigning the first label to the first position. Through permutation operations, we ensure an equal likelihood of each label appearing in any position. Thus, we gather and analyze the labels predicted by GPT-4 in each position. The statistical results of label assignment percentages for each position are presented in Figure 5. Specifically, Figure 5a depicts the evaluation based on opinion resources generated from GPT-4, while Figure 5b pertains to those from FinCaKGs. In Figure 5a, it is evident that GPT-4 tends to predict the *P2-Accountant* label in the first two positions but not the last position. In comparison, Figure 5b, derived from FinCaKGs resources, demonstrates greater balance in the first two positions but overlooks the *P1-CEO* label in the last position. Our conclusion suggests that, while ChatGPT displays positional bias in classification tasks across different opinion resources, this bias is particularly pronounced in resources generated by ChatGPT itself.

7. Case Study II: FinCaKGs support Text Generation

To explore the efficacy of leveraging FinCaKGs in enhancing text generation, we first incorporate FinCaKGs as prior knowledge prior to initiating the generation task. Through comparative analysis between generated text with and without the integration of FinCaKGs, we assess factors including average sentence length, depth of comprehension, and readability. Furthermore, we investigate the impacts of size requirements during text generation to provide a holistic assessment.

Following the financial roles outlined in Section 6, we proceeded with experiments pertaining to each role. However, to mitigate the varied perspectives' im-

Table 10: The readability assessment of the generated text with and without FinCaKGs.

Length	# words/sent		Automated Readability Index (ARI)		Flesch Reading Ease (FRE)	
	w/	w/o	w/	w/o	w/	w/o
50	**33.0±28.6**	16.9±1.0	**23.2±16.9**	13.2±1.7	21.1±38.3	**37.4±8.9**
100	17.3±1.3	**17.7±2.4**	**13.5±2.4**	11.7±2.0	39.8±6.1	**53.5±10.4**
500	**20.0±2.6**	18.2±2.6	**11.7±0.4**	11.1±1.2	54.0±2.2	**55.8±2.7**

Notes: Higher ARI value indicates more educative years to understand; Higher FRE value means much easier to read.

pacts, we computed an average score. In the context of injecting prior knowledge, we provided GPT-4 with a single FinCaKG in triples format as the system content during prompting, alongside a designed question in the user content. Conversely, when prior knowledge was not presented, we assigned a specific financial role to GPT-4 and prompted it to generate responses to the same designed question. Furthermore, we imposed size limits on the generated text to simulate various query scenarios: 50 tokens for a paragraph, 100 tokens for a snippet, and 500 tokens for a report. For evaluating complexity and readability, we utilized two metrics: 1). Automated Readability Index (ARI) [28], which gauges a text's readability based on linguistic features. Higher scores indicate a higher grade level required for comprehension, thus reflecting greater complexity; 2). Flesch Reading Ease (FRE) [29], derived from the mean syllable count per word and the mean word count per sentence. A higher FRE score suggests easier readability, with the maximum score being 100.

The experiment results, detailed in Table 10, exhibit averaged values and standard deviations across three distinct financial roles. Notably, in the '#words/sent' column, it is evident that the generation with FinCaKG tends to produce longer sentences. Moreover, these outputs exhibit higher ARI scores and lower FRE scores, indicating increased complexity and decreased readability when FinCaKG knowledge is utilized. This trend is also noticeable in the paragraph generation scenario (50 tokens) but a bit less evident for longer length scenarios. Furthermore, the high standard deviation values in the paragraph generation scenario suggest instability in the outputs generated with FinCaKGs. Regrettably, the utilization of FinCaKGs as prior knowledge in paragraph generation is not recommended. However, it effectively addresses the issue of flat and monotonous content in longer generation scenarios. For example, in longer length scenarios, particularly evident in the absence of FinCaKGs, there is a substantial decrease in ARI scores. Yet, with the incorporation of FinCaKGs, the complexity score, as measured by ARI, improves from 11 to 13 in a 100 tokens scenario. This enhancement is also observed in the report generation scenario (500 tokens), although not as prominently as in the aforementioned case. To summarize, FinCaKGs demonstrate efficacy in facilitating text generation for longer contexts, but they do not effectively mitigate complexity in shorter scenarios.

8. Limitations

The scope of FinCaKGs is confined to finance-related concepts, thereby excluding coverage of certain significant entities, such as Covid pandemic or energy crises in news events. We could try to diversify the corpus to include a broader range of topics beyond finance. As for causal chains in FinCaKGs, the diminishing informativeness along chains is particularly evident with larger causal leaps. The refinement algorithm to prioritize more direct and causally relevant connections would be practical for improving the coherence and relevance of the generated knowledge graphs. For the evaluation of text generation, we apply with quantitative metrics, which might overlooks valuable qualitative insights that human annotation could provide. We believe that incorporating both assessments can offer a more comprehensive understanding of FinCaKGs' practical applicability in text generation tasks.

9. Conclusion

In this paper, we present a framework to automatically construct the causality knowledge graph from financial reports. We collect the causal and effect span pairs from sentences by implementing a KG-embedded BERT model. To strengthen causality linkage in a graph structure, we propose to transform the pairs of causal spans into structural relations. To facilitate the straightforward visualization of FinCaKG, we provide screenshots to show the properties of nodes and relations and present the cause-effect paths of keywords for expertise discovery. In addition to the original FinCaKG version, we even implemented other related corpus and generated other two versions of FinCaKG. The comparison between distinct versions demonstrates that the FinCaKG framework is capable of uncovering different expertise perspectives based on the diverse cause-effect logic chains. Based on the findings from the two case studies, it was observed that FinCaKGs are capable to represent distinct opinions for diverse financial roles comparing to ChatGPT. Additionally, in the context of text generation tasks, it was noted that incorporating FinCaKGs proved beneficial in addressing the monotony issues in long text generation scenarios.

In the future, we would like to focus on the impact of temporal factors on causality discovery. Meanwhile, based on the current knowledge graph, we are also interested in causality vanishing issues along causal chains and try to detect to which extent we should cut off the connection path, to provide concrete causality chains.

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